

Options trading profits from correlation forecasts

JAMES CHONG

Department of Finance, Real Estate, and Insurance, California State University,
Northridge, 18111 Nordhoff Street, Northridge, CA 91330-8379, USA

E-mail: james.chong@csun.edu

This study examines the profitability of trading currency straddles on the basis of the volatility and correlation forecasts derived from various statistical models. There is evidence to demonstrate that for maximum wealth accumulation, a trader should employ sophisticated models like the exponential GARCH for correlation forecasts and simpler ones like the exponential weighted moving average for volatility forecasts. With differing transaction costs structure between traders, the directional bets taken by the models of the market maker for the most part appear successful, reaping large positive returns. This is especially evident for GBP/DEM straddles and to a lesser extent for JPY/DEM straddles. However, the options trading strategy profits of the price taker are insufficient to outweigh transaction costs, a result considered consistent with market efficiency.

I. INTRODUCTION

In the past decade, the issue of forecasting volatility has witnessed immense interest and has been investigated exhaustively by academics (see, for example, Andersen *et al.*, 2002; Bollerslev *et al.*, 1994 and the references therein). Interest in correlation forecasting is relatively more recent, but developing rapidly, evidenced by the increasing literature on the latter subject in recent years (Siegel, 1997; Campa and Chang, 1998; Gibson and Boyer, 1998; Walter and Lopez, 2000, to name but a few).

The desirability of producing such forecasts is in their wide applicability in the financial arena. Thus far, correlation and volatility forecasts have mostly been evaluated with statistical (e.g. root mean squared forecast error) or hedging (e.g. volatility reduction of hedged over a no-hedge portfolio) criteria.¹ Another criterion that could be employed to appraise the forecasts' effectiveness is by an investigation of the profitability of trading strategies

formed from the forecasts. The success of a model's forecast is in its ability to determine the directional bet required to earn profits in excess of the assumed transaction costs. Some studies have already been carried out along these lines, predominantly centred on trading profits from volatility forecasts in an actual market or correlation forecasts in a simulated market but not on trading profits from correlation forecasts in an actual market. For example, Harvey and Whaley (1992), in analysing the efficiency of the Standard & Poor's (S&P) 100 index options market, employed an implied volatility measure and found that although volatility changes were predictable, abnormal returns were not possible in a trading strategy that took into account transaction costs. Each day, one pair of near-the-money call and put options was employed and delta hedged using S&P 500 futures. Instead of trading different call and put options separately, Noh *et al.* (1994) used a near-the-money straddle to test the efficiency of the S&P 500 index European options market. The advantage

¹ There are other economic loss functions with which to evaluate forecast of second moments, even though they do not focus directly on correlation forecasts. For example, West *et al.* (1993) propose the use of a utility based loss function; Lopez (2001) uses probability forecasts of economic events to evaluate volatility models' forecasts; Christoffersen and Jacobs (2004) examine competing volatility forecasts for option pricing purposes.

of straddle trading is the natural hedge provided to the holder since the call and put positions offset each other. The authors found that forecasting volatility with the GARCH model performed better than the implied volatility regression method and that the former also generates abnormally high rates of return in the S&P 500 index options market.

Several studies have also been conducted in the related area of profits earned by agents trading in a simulated market. Engle *et al.* (1993) proposed a framework for comparing profits generated by different volatility forecasting algorithms for options trading in a simulated market. The authors found that the cumulative profits earned by using a GARCH forecast model with one-day options on the NYSE index dominated those earned by alternative models. Engle *et al.* (1997) extended the framework in analysing long-term options and once again, the GARCH model prevailed. Thus far, the above studies lie within the realm of volatility forecasting. Gibson and Boyer (1998) developed the methodology of Engle *et al.* (1993) to include correlation forecasts in a bivariate setting by comparing GARCH models, Markov switching models and simple moving averages on different simulated stock rainbow option.² It was found that the more sophisticated models (GARCH and Markov switching) produced statistically significantly better forecasts of correlation.

This study strives to contribute to the existing literature by examining the correlation and volatility forecasting performances of various popular GARCH and other models, building upon the series of studies initiated by Harvey and Whaley (1992) with the added feature of correlation forecast evaluation on a wide range of commonly employed models in the liquid OTC currency options market. This study also complements the papers by Gibson and Boyer (1998) since market quoted options are examined instead of being simulated, thus enabling one to evaluate the genuine profitability of trading rules derived from the model forecasts. The research presented could be viewed as an accompaniment to Brooks and Chong (2001), with the same forecasting models, data and time horizon across differing evaluation criterion.

The remainder of this paper is organized as follows. Section I describes the methodology and data employed, while the empirical results on the profitability of the trading follow in Section II. Section III concludes.

II. METHODOLOGY

Data

The data employed for this study are for spot currencies and currency options (specifically at-the-money currency straddles) and are provided by a London-based bank dealing in these instruments, collected at 16:00 GMT. This reduces the errors resulting from nonsynchronous prices. The observation period is 19 July 1993 to 13 September 1999, a length of time slightly more than six years and a total of 1605 daily observations. The option data consists of OTC price quotes for USD/DEM (interpreted as DEM per USD), USD/JPY, DEM/JPY, GBP/USD, and GBP/DEM. These currency options are traded inter-bank, and the price quotes refer to an at-the-money forward straddle, a combination of one European call option and one European put option at the same exercise price equalling the forward rate. An advantage of using OTC straddles is that they are at-the-money, whereas exchange listed straddles are at best nearest-to-the-money. Furthermore, no delta hedging is necessary since the call and put options offset each other.

Computation of volatility/correlation

Employing various forecasting models on a rolling window of 1,000 observations as in-sample data, one-day ahead volatility and correlation are predicted for the out-of-sample period 21 May 1997 to 8 June 1999 (534 data points). The forecasting models – the simple historical, EWMA, GARCH, EGARCH, GJR,³ GARCH-*t*, VEC and BEKK – examined are not new and hence their descriptions are brief (see Table 1). Studies on correlation forecasting, such as Campa and Chang (1998), and Walter and Lopez (2000), tend to exclude the univariate GARCH models, without justifications, focusing their efforts on the multivariate GARCH models. Due to the simplicity of the univariate GARCH models relative to their multivariate counterparts, and their effectiveness at forecasting in the volatility arena, these models are immensely popular among academics and practitioners alike. As such, as long as the conditional correlations predicted by the models are within the bounds of (1, -1), the univariate GARCH models should be included for analysis. Readers are referred to Brooks and Chong (2001), and the references therein, for more details.

² Rainbow options are financial derivatives whose payoff depends on two asset prices, e.g. call spread, outperformance option, call worst, and basket call. The payoffs are not market quoted but rather derived via Monte Carlo simulation.

³ The EGARCH and GJR models are commonly employed to capture the asymmetric response of volatility to positive and negative shocks. In the case of equity returns, such asymmetries typically are attributed to leverage effects, whereby a fall in the value of a firm's stock causes the firm's debt-to-equity ratio to rise. Although such asymmetries in currency returns cannot be attributed to changing leverage, there is equally no reason to suppose that such asymmetries do not exist (McKenzie, 2002).

Table 1. Description of forecasting models

Model	Conditional (co) variance equations for model	Equation
Simple historical	Covariance: $h(x, y)_{t+1} = 1/n \sum_{i=0}^n x_{t-i} y_{t-i}$ Variance: $h(x)_{t+1} = 1/n \sum_{i=0}^n x_{t-i}^2$, $h(y)_{t+1} = 1/n \sum_{i=0}^n y_{t-i}^2$	1
EWMA	Covariance: $h(x, y)_{t+1} = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i x_{t-i} y_{t-i}$ Variance: $h(x)_{t+1} = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i x_{t-i}^2$, $h(y)_{t+1} = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i y_{t-i}^2$	2
GARCH(1,1)	$x_t = \mu + \varepsilon_t$, $\varepsilon_t \sim N(0, h_t)$ $h_t = \gamma + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}$ subject to $\gamma > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$	3
EGARCH(1,1)	$\ln h_t = \gamma + \beta \ln h_{t-1} + \alpha z_{t-1} + \varphi(z_{t-1} - \sqrt{2/\pi})$, where $z_t = \varepsilon_t / \sqrt{h_t}$, and $z_t \sim \text{NID}(0, 1)$	4
GJR(1,1)	$h_t = \gamma + \alpha \varepsilon_{t-1}^2 + \omega \varepsilon_{t-1}^2 S_{t-1}^- + \beta h_{t-1}$ $S_{t-1}^- = 1$ for $\varepsilon_{t-1} < 0$ and 0 otherwise, and subject to $\gamma > 0, \alpha, \beta \geq 0, \alpha + \omega \geq 0, \alpha + \beta + \omega < 1$	5
GARCH(1,1)- t	$x_t = \mu + \varepsilon_t$, $\varepsilon_t \sim t_v(0, h_t)$ $h_t = \gamma + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}$ subject to $\gamma > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$	6
Conditional covariance for models (3)–(6)	$h(x, y)_t = (h(x)_t + h(y)_t - h(x - y)_t)/2$	7
VECH	$\text{vech}(H_t) = W + A \text{vech}(H_{t-1}) + B \text{vech}(\Xi_{t-1} \Xi'_{t-1})$ $\Xi_t \psi_{t-1} \sim N(0, H_t)$	8
BEKK	$H_t = W'W + A'H_{t-1}A + B'\Xi_{t-1}\Xi'_{t-1}B$	9

Note: The acronyms for the models are: EWMA – exponentially weighted moving average; EGARCH – exponential GARCH; GJR – named after Glosten, Jaganathan, and Runkle; GARCH- t – a GARCH developed with a conditional t -distribution; VECH – named after the column-stacking operator unique to this model; BEKK – named after Baba, Engle, Kraft, and Kroner. λ in Equation 2 is assumed to be 0.94 while the parameter estimates for Equations 3–6, 8 and 9 are estimated by maximum likelihood using the algorithm of BFGS. x and y refer to the daily logarithmic returns for assets X and Y respectively, while x_t is x at time t . h_t (and the matrix H_t) refers to the conditional (co)variance at time t . In matrix form, A , B , and W are coefficients, and Ξ is the error term. ψ_{t-1} is the information set at time $t-1$.

Trading strategy rate of return calculation

The trading strategies examined involve JPY/DEM and GPB/DEM and assume an investment of DEM 100 worth of straddles are bought or sold. On each day, each agent applies a forecasting model i in estimating the volatility of tomorrow and hence effectively also forecasting the straddle price of tomorrow. If the straddle price forecast is greater (less) than the closing straddle market price for today, then the straddle is bought (sold). Thus, the rate of return on buying and selling straddles will be

$$\hat{R}_{i,t+1} = R_{t+1}(h_{i,t+1})$$

$$= \begin{cases} \frac{100}{V_{M,t}}(V_{M,t+1} - V_{M,t}) & \text{if bought straddle} \\ -\frac{100}{V_{M,t}}(V_{M,t+1} - V_{M,t}) + 200r_{t+1}^d & \text{if sold straddle} \end{cases}, \quad (10)$$

where $V_{M,t}$ and $V_{M,t+1}$ are respectively the market prices of the straddle at time t and $t+1$, $h_{i,t+1}$ is the conditional variance forecast of model i for $t+1$, r_{t+1}^d is the domestic interest rate at time $t+1$ (where the proceeds of DEM 100 and the initial DEM 100 are invested in a risk free asset

when a straddle is sold). Net of transaction costs, the rate of return from trading straddles is

$$N\hat{R}_{i,t+1} = \hat{R}_{i,t+1} - 100 \left(\frac{\delta_{t+1}}{V_{M,t+1}} + \frac{\delta_t}{V_{M,t+1}} \right), \quad (11)$$

where δ_t and δ_{t+1} are the transaction costs (being half the bid-ask spread and commissions) as a percentage of straddle prices at t and $t+1$ respectively (refer to the section on transaction costs below).

Option pricing model

The pricing model for currency options commonly employed is the Garman-Kohlhagen model (1983), in which a variant is introduced to take into account conditional volatility, and which is represented by

$$C_{i,t+1} = S_{t+1} e^{-r_{t+1}^f \tau} N(d_1) - K_{t+1} e^{-r_{t+1}^d \tau} N(d_2)$$

$$P_{i,t+1} = K_{t+1} e^{-r_{t+1}^d \tau} N(-d_2) - S_{t+1} e^{-r_{t+1}^f \tau} N(-d_1)$$

$$d_1 = \frac{\ln(S_{t+1}/K_{t+1}) + (r_{t+1}^d - r_{t+1}^f + h_{i,t+1}/2)\tau}{\sqrt{h_{i,t+1}\tau}} \quad (12)$$

$$d_2 = d_1 - \sqrt{h_{i,t+1}\tau}$$

where $C_{i,t+1}$ and $P_{i,t+1}$ are the Garman-Kohlhagen call and put currency option price forecasts of model i for $t+1$, S_{t+1} is the market closing spot price (American quotation convention), K_{t+1} is the exercise price (American quotation convention), τ is the time-to-maturity, r_{t+1}^d and r_{t+1}^f are the domestic and foreign risk free interest rates proxy by the one-month Euro deposit rates. Since an at-the-money forward straddle is just a combination of an at-the-money call option and an at-the-money put option, $V_{i,t+1} = C_{i,t+1} + P_{i,t+1}$, with the forward rate $F_{t+1} = S_{t+1} \times e^{(r_{t+1}^d - r_{t+1}^f)\tau} = K_{t+1}$, and τ being 1/365 of a year. Despite the shortcomings of the Garman-Kohlhagen model, emphasis on a pricing model that will explicitly account for stochastic volatility (e.g. the Hull and White model, 1987) is not necessary as the time-to-maturity involves only a day and any error introduced by the use of the Garman-Kohlhagen model will be minimal.

In Equation 12, the volatility forecasting abilities of the various models are under review since the variable of interest is $h_{i,t+1}$. To evaluate the models' correlation forecasts, $h_{i,t+1}$ would be represented by

$$h_{i,t+1}(x-y) = \sigma_{M,t+1}^2(x) + \sigma_{M,t+1}^2(y) - 2\sigma_{M,t+1}(x)\sigma_{M,t+1}(y)\rho_{i,t+1}(x,y), \quad (13)$$

where $\sigma_{M,t+1}^2(x)$ and $\sigma_{M,t+1}^2(y)$ are the quoted implied variances of the X and Y returns, x and y , respectively at time $t+1$, and $\rho_{i,t+1}(x,y)$ is the correlation forecast between x and y predicted by model i for $t+1$. So if the straddle of the cross currency (represented by the variable $x-y$) JPY/DEM is to be priced, isolating the effects of the correlation forecast, the $t+1$ implied volatility quotes for DEM/USD and JPY/USD straddles as well as the correlation forecast between the returns of DEM/USD and JPY/USD are utilized.⁴ Gibson and Boyer's (1998) study of rainbow options in a simulated environment, underlined the differing abilities of models in predicting volatility and correlation. The methodology employed consisted of a package of call and rainbow options, buying (selling) call options will increase (decrease) the sensitivity of the package price to volatility without affecting its sensitivity to correlation, thus increasing (decreasing) the relative importance of volatility. The author will instead employ respectively Equation 12 and the straightforward Equation 13 in assessing the models' volatility or correlation forecasting abilities with trading profits as the evaluation criterion.

Transaction costs

To demonstrate that the straddle trading strategy may lead to abnormal returns, the effect of transaction costs should be included. In this simplest of trading strategies, only basic transaction costs in commissions and quoted bid-ask spreads are considered.⁵ In the OTC currency option market, a conservative commission rate for institutional investors is 0.02%.⁶ The OTC bid-ask spread is much wider, ranging from a maximum of 5.35 (8.06)% and a minimum of 1.35 (3.70)% for the JPY/DEM (GBP/DEM) straddle. Since the transaction prices are equally likely to be at the bid or the ask levels, it is reasonable to assume that, on average, half the bid-ask spread is incurred each time a transaction takes place (Harvey and Whaley, 1992). The average transaction costs as a percentage of the JPY/DEM and the GBP/DEM straddles are 3.72% and 5.73% respectively. Subsequently, a filtering strategy is undertaken so that trades occur only when the straddle price changes in excess of 4% and 6% (for JPY/DEM and GBP/DEM respectively). Naturally, the number of transactions is reduced, with the idle money invested in a risk free deposit. Filtering strategies are practical in that trades are only executed if the forecast price deviation is perceived to be greater than the transaction costs.

With only closing implied volatility available, it could be assumed that the closing volatility is at the bid. Adding 0.5% to the bid volatility will give the ask volatility. A volatility spread of 0.5%⁷ is common for liquid OTC straddles such as those involving JPY/DEM and GBP/DEM. Inputting the bid and ask volatilities into Equation 12 will result in the corresponding bid and ask straddle prices, where the bid-ask spread as a percentage of the straddle *bid* price is determined. In fact, assuming that the closing implied volatility is at the ask and then proceeding with the computation of the bid-ask spread as a percentage of the *ask* straddle price will also produce the same answer.⁸

III. RESULTS

The trading simulations are performed for two types of agents: price takers and market makers, who possess very different transaction costs structure. For price takers, the combined costs of commission and bid-ask spread are

⁴ Thus far, no distinction has been made between the volatility of returns for X and $1/X$ since the squared numerator of the variance formula has rendered them the same.

⁵ Besides quoted bid-ask spreads, other implicit costs include effective bid-ask spreads, price impact costs and opportunity costs. For an overview of institutional trading costs, see Keim and Madhavan (1998).

⁶ We are grateful to Charles Brobst for highlighting this.

⁷ Verified in Bloomberg.

⁸ Empirically, it can also be shown that the volatility spread (0.5) as a percentage of the bid (ask) volatility equates to the corresponding straddle spread as a percentage of the bid (ask) straddle price. We thank Peter Corvi for referring us to Brenner and Subrahmanyam (1994) for the mathematical proof.

Table 2. Rate of return from trading DEM100 worth of straddles per day with JPY/DEM volatility forecasts

	VECH	BEKK	GARCH	EGARCH	GJR	GARCH- <i>t</i>	EWMA	Simple Historical
Trading JPY/DEM with volatility forecasts (out-of-sample period: 21/5/97 – 8/6/99)								
Before transaction costs								
No filter								
Annualized return (%)	–46.87	–36.12	–49.16	–24.99	–62.07	32.29	107.70	–56.25
Rank	5	4	6	3	8	2	1	7
Type of trade as a percent age of total:								
Buy	25.66%	26.03%	33.71%	29.96%	30.90%	37.45%	51.31%	3.00%
No trades	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Sell	74.34%	73.97%	66.29%	70.04%	69.10%	62.55%	48.69%	97.00%
Filter of 4%								
Annualized return (%)	–39.79	–42.62	–50.02	–15.76	–52.53	12.91	88.78	–53.21
Rank	4	5	6	3	7	2	1	8
Type of trade as a percentage of total:								
Buy	21.54%	21.16%	27.90%	24.34%	24.34%	29.03%	42.32%	2.62%
No trade	14.42%	11.99%	13.67%	13.48%	12.36%	21.16%	17.23%	2.81%
Sell	64.04%	66.85%	58.43%	62.17%	63.30%	49.81%	40.45%	94.57%
After transaction costs – market maker								
No filter								
Annualized return (%)	–49.59	–39.39	–51.75	–28.82	–64.01	25.56	97.15	–58.49
Rank	5	4	6	3	8	2	1	7
Filter of 4%								
Annualized return (%)	–42.43	–45.21	–52.24	–19.49	–54.66	8.35	80.80	–55.54
Rank	4	5	6	3	7	2	1	8
After transaction costs – price taker								
No filter								
Annualized return (%)	–100.00	–100.00	–100.00	–100.00	–100.00	–99.99	–99.99	–100.00
Filter of 4%								
Annualized return (%)	–99.99	–99.99	–99.99	–99.99	–99.99	–99.95	–99.94	–100.00

taken into account while only the commission cost is relevant for market makers.

Profits from volatility trading

First, an examination of the profits generated by the volatility forecasting models for OTC JPY/DEM and GBP/DEM straddles is conducted, with the results compared to similar studies (though of different instruments) by Harvey and Whaley (1992) and Noh *et al.* (1994).

JPY/DEM

Table 2 presents the annualized return generated from straddles trading using JPY/DEM volatility forecasts, with and without transaction costs, for both market maker and price taker. Without transaction costs, EWMA is best at predicting the directional change of volatility, hence producing the largest annualized return over all filters. The performance of EWMA is followed

by GARCH-*t* and EGARCH. Negative returns are also not an uncommon sight. The main cause for the models' dismal performances is due to their inability in predicting the volatility change of 7 October 1998 and 8 October 1998,⁹ where a wrong prediction will reduce profits by 38.60% and 66.63% respectively. A reason in part for EWMA's exceptional performance is its successful forecast of such a change on 7 October 1998 (contrast the volatility forecasts between EWMA and GARCH-*t* in Fig. 1). Prior to those two fateful days, the models performed well and registered healthy returns. This result confirms previous findings that, with no transaction costs, GARCH models do have predictive power and hence could be employed as tools to profitable trading strategies.

The introduction of filters complicates the analysis of the models' performances since using higher filters need not result in higher profitability as this is based on the assumption that the prediction of volatility is accurate at higher filters. For example, consider the performance of EWMA under the no filter and 4% filter. One would expect the

⁹ In the week of 5 October 1998, the US dollar fell by about 15% against the yen within three days, the largest one day move in the USD/JPY since the collapse of the Bretton Woods system (International Monetary Fund, 1999).

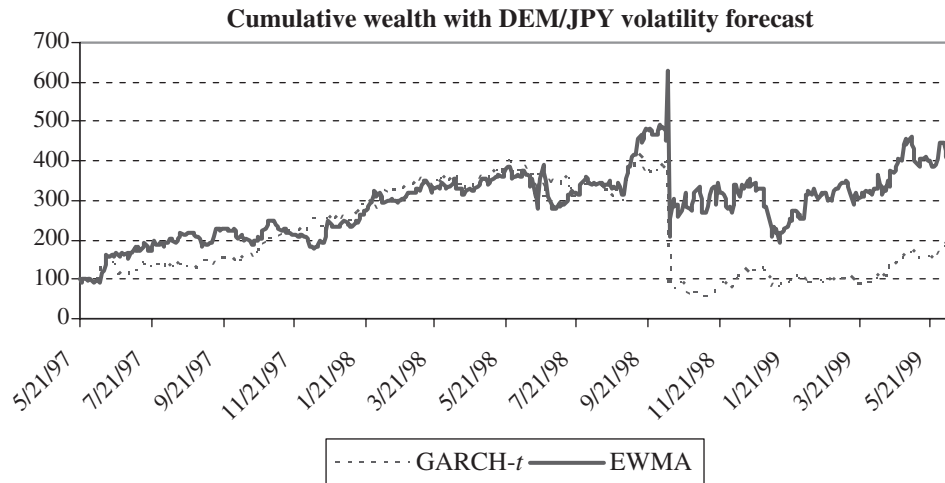


Fig. 1. Cumulative wealth from straddles trading with JPY/DEM volatility forecasts

Table 3. Rate of return from trading DEM100 worth of straddles per day with GBP/DEM volatility forecasts

	VECH	BEKK	GARCH	EGARCH	GJR	GARCH- <i>t</i>	EWMA	Simple historical
Trading GBP/DEM with volatility forecasts (out-of-sample period: 21/5/97 – 8/6/99)								
Before transaction costs								
No filter								
Annualized return (%)	25.40	31.22	90.54	38.59	79.88	95.29	9.69	75.92
Rank	7	6	2	5	3	1	8	4
Type of trade as a percentage of total:								
Buy	28.09%	32.77%	22.28%	34.83%	27.53%	26.59%	34.83%	40.45%
No trade	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Sell	71.91%	67.23%	77.72%	65.17%	72.47%	73.41%	65.17%	59.55%
Filter of 6%								
Annualized return (%)	56.29	5.73	31.13	29.66	68.72	17.34	−5.55	75.63
Rank	3	7	4	5	2	6	8	1
Type of trade as a percentage of total:								
Buy	15.92%	17.42%	13.30%	15.92%	11.99%	13.86%	20.60%	18.16%
No trade	27.53%	36.33%	34.64%	35.02%	36.52%	37.45%	29.59%	30.90%
Sell	56.55%	46.25%	52.06%	49.06%	51.50%	48.69%	49.81%	50.94%
After transaction costs – market maker								
No filter								
Annualized return (%)	19.04	24.57	80.90	31.57	70.78	85.41	4.13	67.01
Rank	7	6	2	5	3	1	8	4
Filter of 6%								
Annualized return (%)	50.52	2.28	26.75	25.35	63.25	13.58	−8.95	69.44
Rank	3	7	4	5	2	6	8	1
After transaction costs – price taker								
No filter								
Annualized return (%)	−100.00	−100.00	−100.00	−100.00	−100.00	−100.00	−100.00	−100.00
Filter of 6%								
Annualized return (%)	−100.00	−99.99	−99.99	−99.99	−99.99	−99.99	−100.00	−99.99

model to perform better under the 4% filter since higher profits will result from correctly predicting the 4% volatility move. However, the annualized return from the 4% filter is lower than that of the no filter, partially attributed by the incorrect prediction of 4% volatility moves and by

the reduction in profits by investing in a risk free instrument instead of profitable but less volatile directional bets. Across models, an example (refer to Table 3) would be to compare VECH with its competitor GARCH-*t*. Under the no filter and without transaction costs scenario, VECH

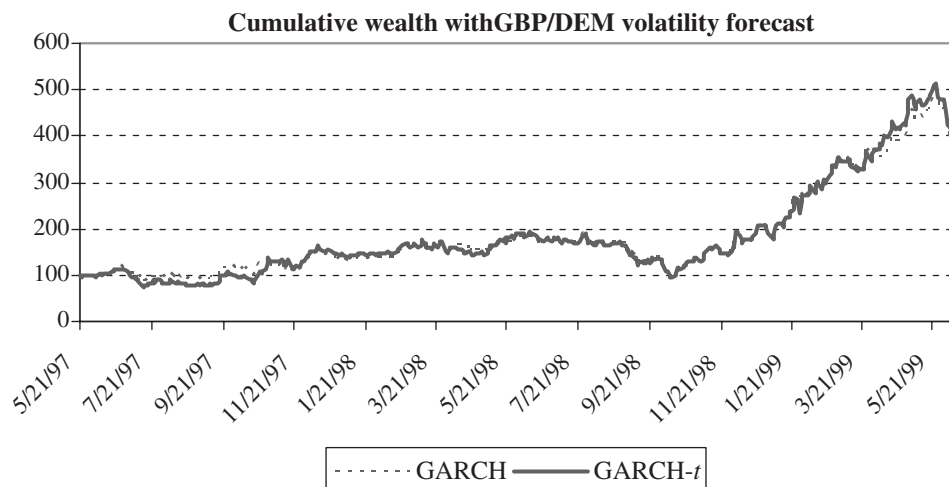


Fig. 2. Cumulative wealth from straddles trading with GBP/DEM volatility forecasts

is placed seventh while GARCH- t came in first. However, VECH is third while GARCH- t is sixth in the 6% filter case because of VECH's superiority in predicting larger volatility swings.

With transaction costs, the price taker basically lost his whole investment. This is not surprising since the average transaction cost is 3.72% per day, with the bid-ask spread as a percentage of the straddle prices comprising the bulk of the cost. In line with previous research, this result is consistent with the efficient market hypothesis that once transaction costs are included in any trading strategy, no profitable venture will materialize. For the market maker, on the other hand, the EWMA and GARCH- t models registered positive annualized returns (with and without the use of filters) since the bid-ask spread is not included as part of his cost structure.

GBP/DEM

The annualized return for GBP/DEM straddles trading, prior to transaction costs is positive across models (refer to Table 3), a contrast from JPY/DEM straddles trading. With a lower GBP/DEM return variance (0.2756 versus 0.8367 for DEM/JPY), the forecasting models are more successful at mimicking the volatility structure of GBP/DEM returns than of JPY/DEM returns, hence producing a more successful trading strategy for GBP/DEM straddles (the consistent upsurge in cumulative wealth from GBP/DEM straddles trading is evident from Fig. 2). Excluding transaction costs, the most dependable model is the GJR, falling within the top three positions for each of the filter levels. This is a sharp distinction from its performance with JPY/DEM straddles trading. GARCH- t ranks first under no filter but its form dips under the 6% filter, reflecting the lack of profitable opportunities with smaller price swings as a result of imposing a higher filter. In this

case, EWMA performs poorly, with fellow laggards in the VECH and BEKK models.

With transaction costs, the price taker once again lost his investment, in light of a hefty average transaction cost of 5.73% per day. For the market maker, all the models registered positive annualized returns, with higher returns in the majority of cases when no filter is imposed than when a 6% filter is applied (the cases are equally divided for JPY/DEM). This is indicative of the forecasting models' failure in predicting large volatility moves, whose strength is in accumulating wealth via less volatile directional bets.

Profits from correlation trading

The following section, executed in the spirit of Gibson and Boyer (1998), will scrutinize the models' correlation forecasting capability, applying the forecast of USD/DEM, USD/JPY correlation for JPY/DEM straddles trading and that of USD/GBP, USD/DEM correlation for GBP/DEM straddles.

USD/DEM, USD/JPY

Table 4 presents the results from trading DEM/JPY straddles with USD/DEM, USD/JPY correlation forecasts, prior to and subsequent to transaction costs. Before transaction costs, only the GJR and BEKK models recorded positive annualized returns. Nonetheless this does not preclude the forecasts from the volatility spikes of 7 and 8 October 1998, hence affecting their performances adversely. Prior to those dates, the returns from straddles trading were strong, as illustrated by GJR and BEKK in Fig. 3. For a filter of 4%, GARCH- t and simple historical are the two best models. This suggests that the strength of GJR and BEKK lie with directional bets on straddle prices of less than 4% while the forte of GARCH- t and simple historical are with price swings of larger than 4%.

Table 4. Rate of return from trading DEM100 worth of straddles per day with USD/DEM, USD/JPY correlation forecasts

	VECH	BEKK	GARCH	EGARCH	GJR	GARCH- <i>t</i>	EWMA	Simple historical
Trading JPY/DEM with USD/DEM, USD/JPY correlation forecasts (Out-of-sample period: 21/5/97 – 8/6/99)								
Before transaction costs								
No filter								
Annualized return (%)	−18.29	4.77	−22.34	−12.89	4.98	−23.09	−27.47	−17.71
Rank	5	2	6	3	1	7	8	4
Type of trade as a percentage of total:								
Buy	47.94%	41.39%	46.44%	36.52%	41.39%	40.26%	55.62%	18.16%
No trade	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Sell	52.06%	58.61%	53.56%	63.48%	58.61%	59.74%	44.38%	81.84%
Filter of 4%								
Annualized return (%)	−8.45	−16.97	−33.81	−16.23	−13.81	13.65	−16.37	−6.24
Rank	3	7	8	5	4	1	6	2
Type of trade as a percentage of total:								
Buy	39.33%	31.46%	40.07%	30.71%	33.33%	33.33%	48.69%	12.92%
No trade	17.23%	18.54%	10.30%	12.92%	14.61%	13.86%	11.80%	12.55%
Sell	43.45%	50.00%	49.63%	56.37%	52.06%	52.81%	39.51%	74.53%
After transaction costs – market maker								
No filter								
Annualized return (%)	−22.46	−0.57	−26.31	−17.33	−0.37	−27.02	−31.17	−21.91
Rank	5	2	6	3	1	7	8	4
Filter of 4%								
Annualized return (%)	−12.34	−20.44	−36.85	−19.97	−17.58	8.64	−20.14	−10.44
Rank	3	7	8	5	4	1	6	2
After transaction costs – price taker								
No filter								
Annualized return (%)	−100.00	−99.99	−100.00	−100.00	−99.99	−100.00	−100.00	−100.00
Filter of 4%								
Annualized return (%)	−99.97	−99.97	−99.99	−99.98	−99.98	−99.98	−99.99	−99.98

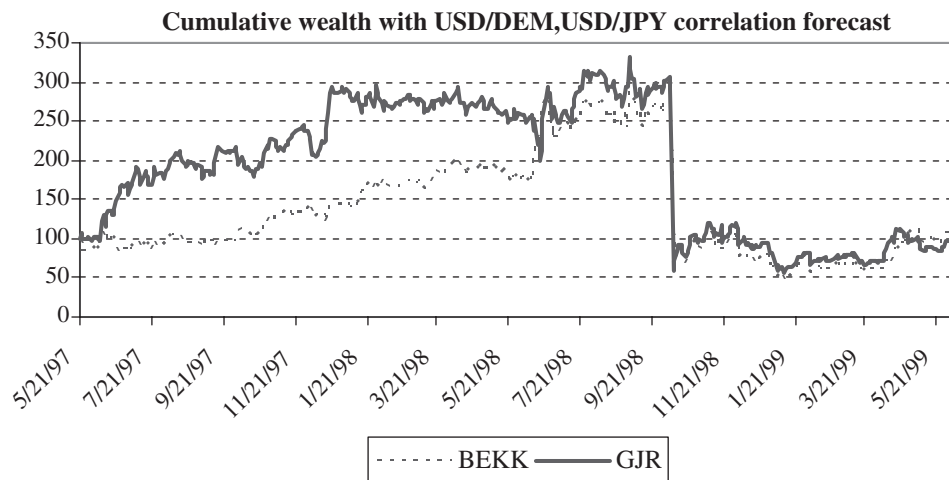


Fig. 3. Cumulative wealth from JPY/DEM straddles trading with USD/DEM, USD/JPY correlation forecasts

As in trading with volatility forecasts, transaction costs or more specifically the bid-ask spread, plays a significant part as it tends to overwhelm the profits generated by the successful trading of models. While EWMA was best with JPY/DEM volatility forecast net

of transaction costs from the perspective of the market maker, GARCH-*t* and GJR are most reliable with USD/DEM, USD/JPY correlation forecast in trading of JPY/DEM straddles, with and without filters respectively.

Table 5. Rate of return from trading DEM100 worth of straddles per day with USD/GBP, USD/DEM correlation forecasts

	VECH	BEKK	GARCH	EGARCH	GJR	GARCH- <i>t</i>	EWMA	Simple Historical
Trading GBP/DEM with USD/GBP, USD/DEM correlation forecasts (out-of-sample period: 21/5/97 – 8/6/99)								
Before transaction costs								
No filter								
Annualized return (%)	100.68	58.90	80.35	164.19	97.07	66.77	87.17	64.48
Rank	2	8	5	1	3	6	4	7
Type of trade as a percentage of total:								
Buy	55.81%	63.11%	66.29%	63.30%	66.10%	57.30%	69.10%	50.00%
No trade	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Sell	44.19%	36.89%	33.71%	36.70%	33.90%	42.70%	30.90%	50.00%
Filter of 6%								
Annualized return (%)	51.12	83.38	56.92	119.31	98.90	101.46	91.95	58.06
Rank	8	5	7	1	3	2	4	6
Type of trade as a percentage of total:								
Buy	41.20%	51.31%	53.37%	48.88%	51.12%	46.07%	55.06%	43.45%
No trade	28.28%	22.66%	24.91%	26.40%	25.09%	25.47%	24.53%	17.79%
Sell	30.52%	26.03%	21.72%	24.72%	23.78%	28.46%	20.41%	38.76%
After transaction costs – market maker								
No filter								
Annualized return (%)	90.53	50.86	71.23	150.84	87.10	58.33	77.70	56.15
Rank	2	8	5	1	3	6	4	7
Filter of 6%								
Annualized return (%)	45.59	76.16	50.92	111.10	91.32	93.82	84.58	51.45
Rank	8	5	7	1	3	2	4	6
After transaction costs – price taker								
No filter								
Annualized return (%)	–100.00	–100.00	–100.00	–100.00	–100.00	–100.00	–100.00	–100.00
Filter of 6%								
Annualized return (%)	–100.00	–100.00	–100.00	–100.00	–100.00	–100.00	–100.00	–100.00

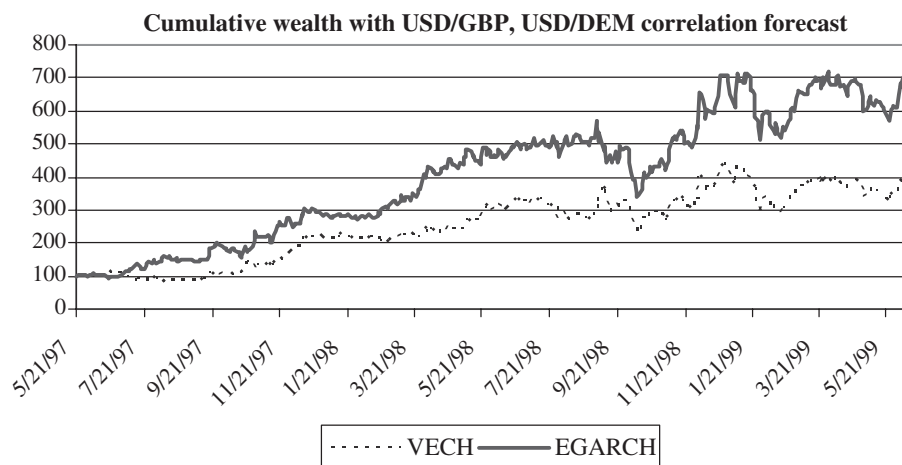


Fig. 4. Cumulative wealth from GBP/DEM straddles trading with USD/GBP, USD/DEM correlation forecasts

USD/GBP, USD/DEM

Prior to transaction costs, positive annualized returns are once again exhibited for the trading of GBP/DEM straddles, this time with just the USD/GBP, USD/DEM

correlation forecasts (see Table 5). EGARCH lead for all filter levels with an impressive 164.19% and 119.31% annualized return respectively for no filter and a 6% filter. The cumulative wealth from the USD/GBP, USD/DEM forecasts (Fig. 4) surpasses that of the USD/DEM,

USD/JPY forecasts (Fig. 3) and this success can be attributed to the stability of the GBP/DEM returns volatility, in contrast to the volatility of the JPY/DEM returns.

From the perspective of a market maker with transaction costs, EGARCH continued its dominance, trailed by VEC and GJR. Analysed jointly with the returns from GBP/DEM volatility forecasts, GJR appears most consistent, appearing in the top three position for different filters, with and without transaction costs.

IV. CONCLUSIONS

Without transaction costs, the directional bets taken by the models for the most part appear successful, reaping positive returns, and corroborating with the findings in Harvey and Whaley (1992) and Noh *et al.* (1994). This is especially evident for GBP/DEM straddles and due to the JPY/DEM volatility spike in October 1998, to a certain extent for JPY/DEM straddles.

As transaction costs play a substantial role in studies on trading profitability, it is important from which market participants' perspective our analyses take since differing conclusions would result. Due to the incurrence of large transaction costs, attributed mainly to the bid-ask spread, the profits generated with the forecasting models are unlikely to be preserved from the standpoint of the price taker. Since profits are not forthcoming after transaction costs, market efficiency holds. On the other hand, the revenue from the bid-ask spread for a market maker would relieve him from such overwhelming costs, thus allowing him to preserve his investment profits.

While the sophisticated forecasting models incorporate more stylized features of asset returns into its structure than the simpler ones, this does not suggest that they are more accurate. Gibson and Boyer (1998) endorse the sophisticated correlation forecasting models in their simulated environment on stock rainbow options, while Byström (2002), in his follow up with currency rainbow options, supports a simpler model like EWMA, which performs equally well with wealth accumulation. In this study, the author advocates employing sophisticated models (e.g. GJR, EGARCH) for correlation forecasts and the simpler ones (e.g. simple historical, EWMA) for volatility forecasts. This augurs well for the simpler models as it signifies possible relevance in certain scenarios.

Despite using daily closing prices, the one aspect shared by the successful forecasting models is in generating overwhelming profits (prior to transaction costs). As such, we could expect better correlation and volatility forecasts,

and hence greater trading profits from using intra-day data. This holds promise for future research especially since daily volatility forecasts constructed from multiple intra-day volatility forecasts result in superior out-of-sample predictions (Martens, 2001).

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